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plane:

Defn: A plane is a surface such that if any two points are taken on it, the straight line joining them lies wholly in the surface.

Corolla: General equation of a given point (x_1, y_1, z_1)

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0$$

c-2: The equation of a plane passing through two points (x_1,y_1,z_1) & (x_2,y_2,z_2)

$$(x-x_1)(y-y_2)-(x-x_2)(y-y_1) = A\{(x-x_1)(z-z_2)-(x-x_2)(z-z_1)\}$$

C-3:- plane passing through three points:

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

Ex-1: Find the equation of the plane passing through the intersection of the planes $x+2y+3z+4=0$ and $4x+3y+2z+1=0$ and the point $(1,2,3)$.

Soln: Any plane through the intersection of two planes is

$$(x+2y+3z+4)+K(4x+3y+2z+1)=0 \quad \ldots (1)$$

Since it passes through $(1,2,3)$,

$$18+17K=0 \Rightarrow K=\frac{-18}{17}$$

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Putting the value in eqn ①,

$$(x+2y+3z+4) - \frac{18}{17}(4x+3y+2z+1) = 0$$

$$\Rightarrow 55x + 20y - 15z - 50 = 0$$

$$\Rightarrow 11x + 4y - 3z = 10$$ ~~Ans~~

Ex-2:

Find the equation of the plane through the point $(4,-2,1)$ and parallel to the plane whose direction ratios are $7, 2, -3$.

Soln:

The equation of the plane the point $(4,-2,1)$,

$$a(x-x_1) + b(y-y_1) + c(z-z_1) = 0$$

$$\Rightarrow a(x-4) + b(y+2) + c(z-1) = 0$$

It is parallel to the plane whose direction ratio's are $a=7$, $b=2$, $c=-3$, Ex-4: Find the equation of the plane passing through the lines of the intersection of the planes $2x-y=0$ and $3z-y=0$ and perpendicular to the plane $4x+5y-3z+1=0$

Soln:

The required plane,

$$(2x-y)+k(3z-y)=0$$

$$\Rightarrow 2x-(1+k)y+3kz=0$$

It is perpendicular to $4x+5y-3z+1=0$

$$\therefore 2.4-(1+k).5+3k(-3)=0$$

$$\Rightarrow k=\frac{-3}{19}$$

The required plane is

$$2x-y+\frac{3}{19}(3z-y)=0$$

$$\therefore 28x-17y+9z=0$$

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It passes through $(1, -2, 3)$

$$(5+6-1) + k(7-9-14) = 0$$

$$\Rightarrow 10 + k(-16) = 0$$

$$\Rightarrow k = \frac{10}{16} = \frac{5}{8}$$

from ④

$$(5x-3y-1) + \frac{5}{8}(7x-3z-19) = 0$$

$$\therefore 25x - 8y - 5z - 26 = 0$$

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$7(x-4)+2(y+2)-3(z-1)=0$

$\Rightarrow 7x-28+2y+4-3z+3=0$

$\Rightarrow 7x+2y-3z=21$ Ans

Ex-3:

Find the equation of the plane through the point (4, 0, 1) and parallel to the plane $4x+3y-12z+6=0$

Solⁿ: The required plane,

$$4x+3y-12z+k=0$$

It passes through the point (4, 0, 1),

$$4.4+3.0-12.1+k=0$$

$\Rightarrow 16-12+k=0$

$\Rightarrow k=-4$

The required plane is $4x+3y-12z-4=0$

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Ex-5:

Find the equation of the plane through the points $(2,2,1)$ and $(9,3,6)$ and perpendicular to the plane $2x+6y+6z=9$

Solⁿ: The required plane is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a & b & c \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-2 & y-2 & z-1 \\ 9-2 & 3-2 & 6-1 \\ 2 & 6 & 6 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-2 & y-2 & z-1 \\ 7 & 1 & 5 \\ 2 & 6 & 6 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)\begin{vmatrix} 1 & 5 \\ 6 & 6 \end{vmatrix} - (y-2)\begin{vmatrix} 7 & 5 \\ 2 & 6 \end{vmatrix} + (z-1)\begin{vmatrix} 7 & 1 \\ 2 & 6 \end{vmatrix} = 0$$

$$\Rightarrow 3x+4y-5z=9$$

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Ex-6:

Find the equation of the plane through the line

$$\frac{x-2}{3} = \frac{y-3}{5} = \frac{z}{7}$$ and passing through the point $(1,-2,3)$.

Soln: $\frac{x-2}{3} = \frac{y-3}{5}$

$\Rightarrow 5x-10 = 3y-9$

$\Rightarrow 5x-3y-1 = 0$

$\frac{x-2}{3} = \frac{z}{7}$

$\Rightarrow 7x-14 = 3z$

$\Rightarrow 7x-3z-14 = 0$

The required plane is

$(5x-3y-1) + k(7x-3z-14) = 0$

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Gradient: Let $\phi(x,y,z)$ be defined and differentiable at each point (x,y,z) in a certain region of space. Then the gradient of ϕ, written $\nabla\phi$ is defined by $\nabla\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)(x,y,z)$

Divergent: Let $V(x,y,z) = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$ be defined and differentiable at each point (x,y,z) in a certain region of space. Then the divergence of v, written $\nabla\cdot\vec{v} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot$

$$(v_1\hat{i} + v_2\hat{j} + v_3\hat{k})\cdot$$

$$= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

Curl: If $v(x,y,z)$ be defined vector field. Then the curl of v, written $\nabla\times v$

$$\nabla\times\vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Ex: Suppose $A = x^2z^2\hat{i} - 2y^2z^2\hat{j} + xy^2z\hat{k}$. Find $\nabla \times A$ at $P(1,-1,1)$

Soln: $A = x^2z^2\hat{i} - 2y^2z^2\hat{j} + xy^2z\hat{k}$

$$\nabla \times A = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2z^2 & -2y^2z^2 & xy^2z \end{vmatrix}$$

$$= \hat{i}\left(2xyz + 4y^2z\right) - \hat{j}\left(y^2z - 2x^2z\right) + \hat{k}\cdot 0$$

$P(1,-1,1)$, $\nabla \times \vec{A} = (-2+4)\hat{i} - \hat{j}(-1)$

$$= 2\hat{i} + \hat{j}$$

Ex: Suppose $\phi(x,y,z) = 3x^2y - y^3z^2$, $\nabla\phi$ at point $(1,-2,-1)$

$$\nabla\phi = \frac{\partial}{\partial x}\left(3x^2y, -y^3z^2\right)\hat{i} + \frac{\partial}{\partial y}\left(3x^2y, -y^3z^2\right)\hat{j} + \frac{\partial}{\partial z}\left(3x^2y, -y^3z^2\right)\hat{k}$$

$$= (6xy - 0)\hat{i} + (3x^2 - 3y^2z^2)\hat{j} - 3y^3z\hat{k}$$

$P(1,-2,-1)$ $\nabla\phi = -12\hat{i} - 9\hat{j} - 24\hat{k}$.

$$\nabla \phi(1,3,1) = (2\cdot 1, 3\cdot 1 - 4\cdot 3\cdot 1)\hat{i} + (1-4\cdot 1\cdot 1)\hat{j} + (3-8\cdot 1\cdot 3\cdot 1)\hat{k}$$

$$= -6\hat{i} - 3\hat{j} - 21\hat{k}$$

The unit vector in the directional derivative,

$$\vec{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{9}} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$$

The required directional derivatives,

$$\vec{a}\cdot\nabla\phi = (-6\hat{i} - 3\hat{j} - 21\hat{k})\cdot\left(\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}\right)$$

$$= -4 + 1 + 14$$

$$= 11$$

\# Show that, $\nabla r^n = n r^{n-2} \vec{r}$

Soln:

L.H.S $= \nabla r^n$

$$= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\left(\sqrt{x^2+y^2+z^2}\right)^n$$

$$= \frac{\partial}{\partial x}(x^2+y^2+z^2)^{n/2}\hat{i} + \frac{\partial}{\partial y}(x^2+y^2+z^2)^{n/2}\hat{j} + \frac{\partial}{\partial z}(x^2+y^2+z^2)^{n/2}\hat{k}$$

$$= \frac{n}{2}(x^2+y^2+z^2)^{n/2-1}\cdot 2x\hat{i} + \frac{n}{2}(x^2+y^2+z^2)^{n/2-1}\cdot 2y\hat{j} + \frac{n}{2}(x^2+y^2+z^2)^{n/2-1}\cdot 2z\hat{k}$$

$$= n(x^2+y^2+z^2)^{n/2-1}(x\hat{i}+y\hat{j}+z\hat{k})$$

$$= n(r^2)^{\frac{n-2}{2}}\cdot\vec{r}$$

$$= n r^{n-2}\cdot\vec{r}$$

$=$ R.H.S

(Ans)

Ex: Let $\phi = x^2yz - 4xyz^2$. Find the directional derivative of ϕ at $P(1,3,1)$ in the direction of $2\hat{i}-\hat{j}-2\hat{k}$.

Ans: we have,

$$\nabla\phi = \frac{\partial}{\partial x}(x^2yz, -4xyz^2)\hat{i} + \frac{\partial}{\partial y}(x^2yz, -4xyz^2)\hat{j} + \frac{\partial}{\partial z}(x^2yz, -4xyz^2)\hat{k}$$

$$= (2xyz, -4yz^2)\hat{i} + (x^2z, -4xz^2)\hat{j} + (x^2y, -8xyz)\hat{k}$$

Ex: Find the gradient, divergence and curl of the vector $x^2z\hat{i} - y^3z^2\hat{j} + xy^2z\hat{k}$.

Soln:

Let the function, $f = x^2z\hat{i} - y^3z^2\hat{j} + xy^2z\hat{k}$

The gradient, $\nabla f = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)(x^2z, -y^3z^2, xy^2z)$

$$= \frac{\partial}{\partial x}(x^2z, -y^3z^2, xy^2z)\hat{i} + \frac{\partial}{\partial y}(x^2z, -y^3z^2, xy^2z)\hat{j} + \frac{\partial}{\partial z}(x^2z, -y^3z^2, xy^2z)\hat{k}$$

$$= (2xz, 0, y^2z)\hat{i} + (0, -3y^2z^2, 2xyz)\hat{j} + (x^2, -2y^3z, xy^2)\hat{k}$$

$$= (2xz + y^2z)\hat{i} + (-3y^2z^2 + 2xyz)\hat{j} + (x^2 - 2y^3z + xy^2)\hat{k}$$

The divergent $\nabla \cdot f = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right) \cdot (x^2z\hat{i} - y^3z^2\hat{j} + xy^2z\hat{k})$

$$= \frac{\partial}{\partial x}(x^2z) + \frac{\partial}{\partial y}(-y^3z^2) + \frac{\partial}{\partial z}(xy^2z)$$

$$= 2xz - 3y^2z^2 + xy^2$$

The curl, $\nabla \times f = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2z & -y^3z^2 & xy^2z \end{vmatrix}$

$$= \hat{i}\left(\frac{\partial}{\partial y}(xy^2z) - \frac{\partial}{\partial z}(-y^3z^2)\right) + \hat{j}\left(\frac{\partial}{\partial x}(xy^2z) - \frac{\partial}{\partial z}(x^2z)\right) + \hat{k}\cdot 0$$

$$= \hat{i}(2xyz + 2y^3z) - \hat{j}(y^2z - x^2)$$ (Ans)